



ESP32-CAM Development Board

SKU:DFR0602

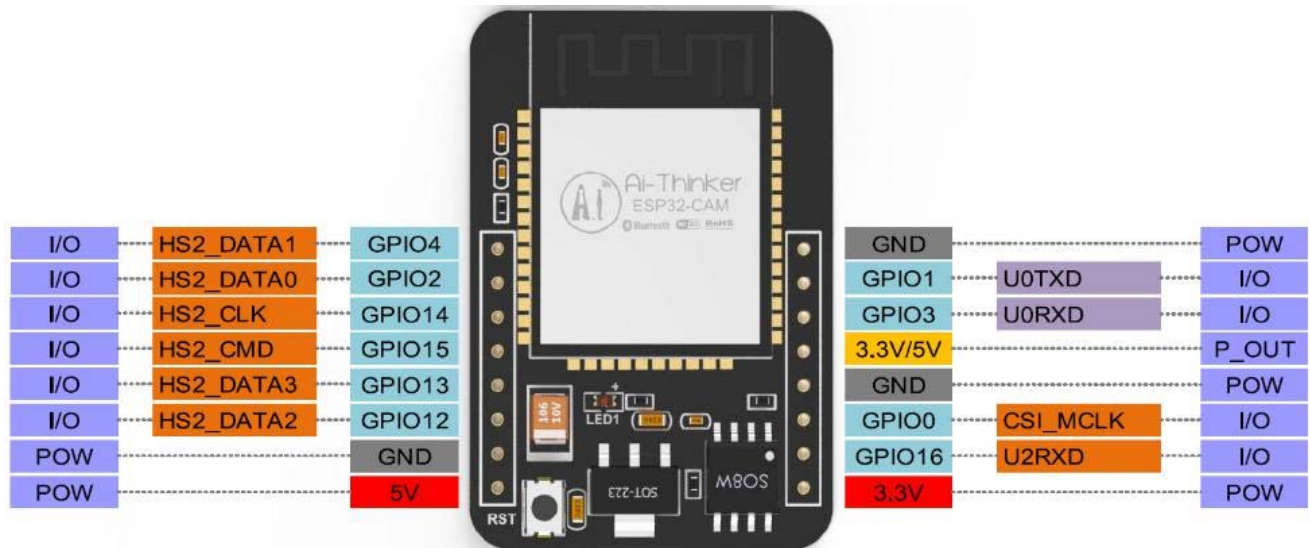
INTRODUCTION

ESP32-CAM is a low-cost ESP32-based development board with onboard camera, small in size. It is an ideal solution for IoT application, prototypes constructions and DIY projects.

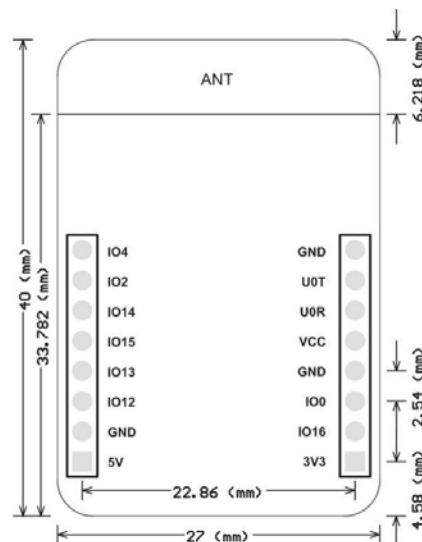
The board integrates WiFi, traditional Bluetooth and low power BLE , with 2 high-performance 32-bit LX6 CPUs. It adopts 7-stage pipeline architecture, on-chip sensor, Hall sensor, temperature sensor and so on, and its main frequency adjustment ranges from 80MHz to 240MHz.

Fully compliant with WiFi 802.11b/g/n/e/i and Bluetooth 4.2 standards, it can be used as a master mode to build an independent network controller, or as a slave to other host MCUs to add networking capabilities to existing devices

ESP32-CAM can be widely used in various IoT applications. It is suitable for home smart devices, industrial wireless control, wireless monitoring, QR wireless identification, wireless positioning system signals and other IoT applications. It is an ideal solution for IoT applications.



Schematic Diagram



Dimension Diagram

Notes:

1. Please be sure that the power supply for the module should be at least 5V 2A, otherwise maybe there would be water ripple appearing on the image.

2.ESP32 GPIO32 pin is used to control the power of the camera, so when the camera is in working, pull GPIO32 pin low.

3.Since IO pin is connected to camera XCLK, it should be left floating in using, and do not connect it to high/low level.

4.The product has been equipped with default firmware before leaving the factory, and we do not provide additional ones for you to download. So, please be cautious when you choose to burn other firmwares.

FEATURES

- Up to 160MHz clock speed, Summary computing power up to 600 DMIPS
- Built-in 520 KB SRAM, external 4MPSRAM
- Supports UART/SPI/I2C/PWM/ADC/DAC
- Support OV2640 and OV7670 cameras, Built-in Flash lamp.
- Support image WiFi upload
- Support TF card
- Supports multiple sleep modes.
- Embedded Lwip and FreeRTOS
- Supports STA/AP/STA+AP operation mode
- Support Smart Config/AirKiss technology
- Support for serial port local and remote firmware upgrades (FOTA)

SPECIFICATION

- SPI Flash: default 32Mbit
- RAM: built-in 520 KB+external 4MPSRAM
- Dimension: 27*40.5*4.5 (± 0.2) mm/1.06*1.59*0.18"
- Bluetooth: Bluetooth 4.2 BR/EDR and BLE standards
- Wi-Fi: 802.11b/g/n/e/i
- Support Interface: UART, SPI, I2C, PWM
- Support TF card: maximum support 4G
- IO port: 9
- Serial Port Baud-rate: Default 115200 bps

- Image Output Format: JPEG(OV2640 support only), BMP, GRAYSCALE
- Spectrum Range: 2412 ~2484MHz
- Antenna: onboard PCB antenna, gain 2dBi
- Transmit Power: 802.11b: 17±2 dBm (@11Mbps);
802.11g: 14±2 dBm (@54Mbps);
802.11n: 13±2 dBm (@MCS7)
- Receiving Sensitivity: CCK, 1 Mbps : -90dBm;
CCK, 11 Mbps: -85dBm;
6 Mbps (1/2 BPSK): -88dBm;
54 Mbps (3/4 64-QAM): -70dBm;
MCS7 (65 Mbps, 72.2 Mbps): -67dBm
- Power consumption: Turn off the flash: 180mA@5V
Turn on the flash and adjust the brightness to the maximum:
310mA@5V
Deep-sleep: the lowest power consumption can reach 6mA@5V
Moderm-sleep: up to 20mA@5V
Light-sleep: up to 6.7mA@5V
- Security: WPA/WPA2/WPA2-Enterprise/WPS
- Power supply range: 5V
- Operating temperature: -20 °C ~ 85 °C
- Sorage environment: -40 °C ~ 90 °C, < 90%RH
- Weight: 10g

SHIPPING LIST

- ESP32-CAM Development Board x1

